

Lab1: Configuration of your PC to carry out the labs

For the Labs (practical work), we will work on a software called IMUNES. It is a network emulator that has the ability to create terminals (PCs or servers), switches and routers in a fairly simple way and provide a command line interface to interact with each emulated machine. We will use a pre-configured virtual machine in which the software is already installed. This virtual machine runs under VirtualBox, free software that you can download for most platforms here:

<https://www.virtualbox.org/wiki/Downloads>

You can download and install the extensions that will allow you to get the most out of this software (see the VirtualBox documentation).

Once you have VirtualBox installed on your machine, you can download the image of the virtual machine that we are going to use, at the following address: [imunes, any version from 10.2 to 14.2](https://imunes.net/download)

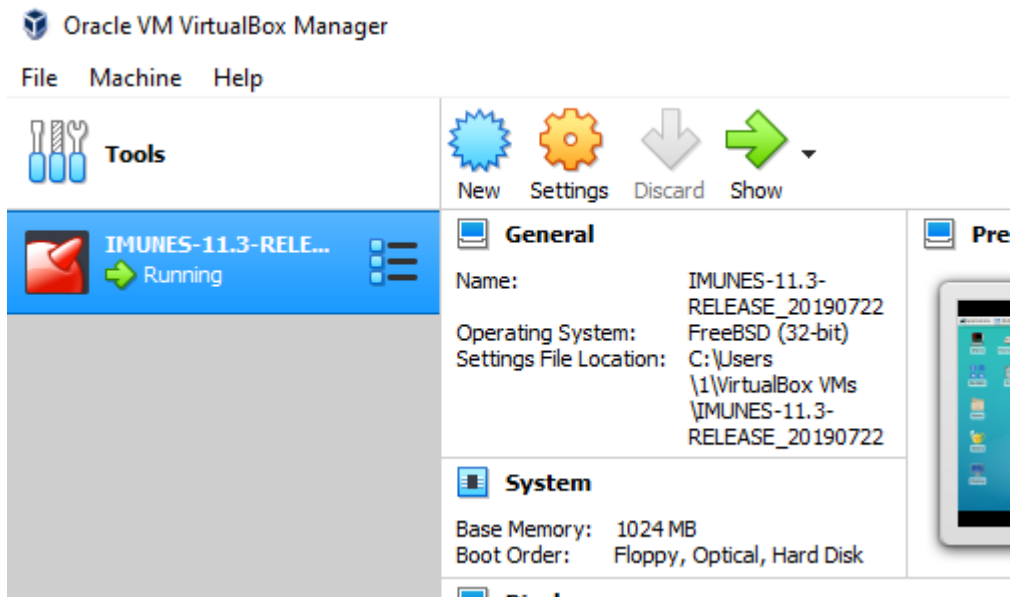
<https://imunes.net/download>

Then double click on the downloaded IMUNES-yourversion-RELEASE.ova file then click on Import as shown in the following figure.

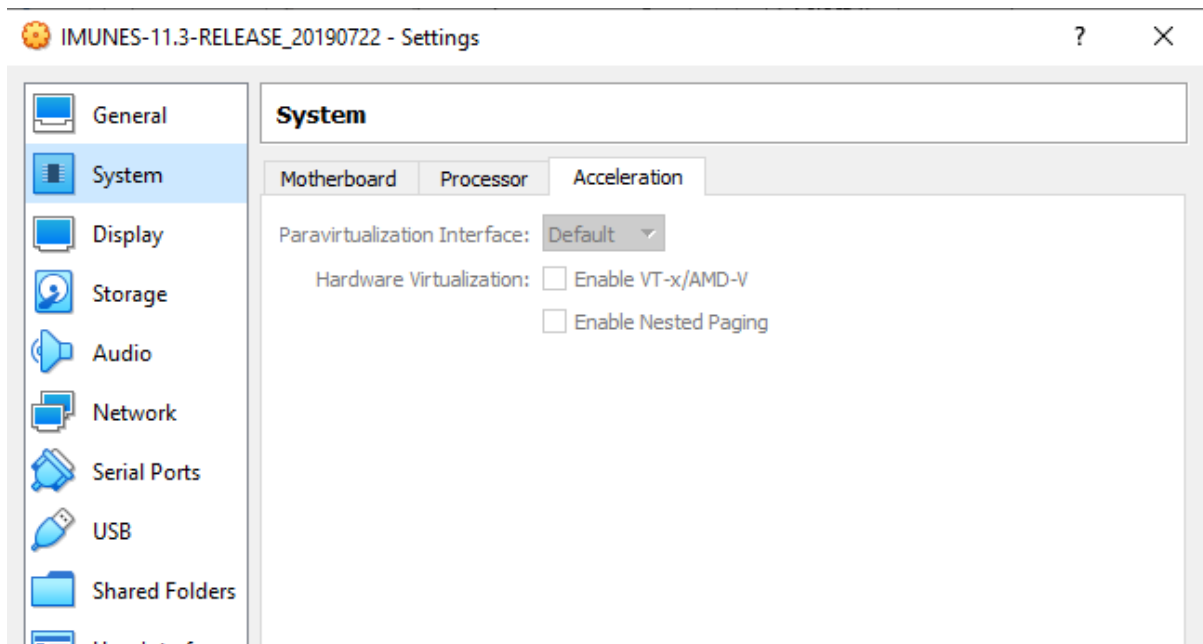


If your get an error,

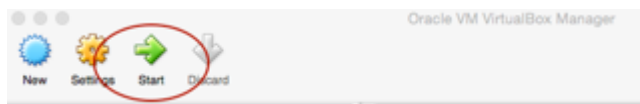
Go to setting



Settings->system->Acceleration and deactivate Enable VT and Enable Nested Paging



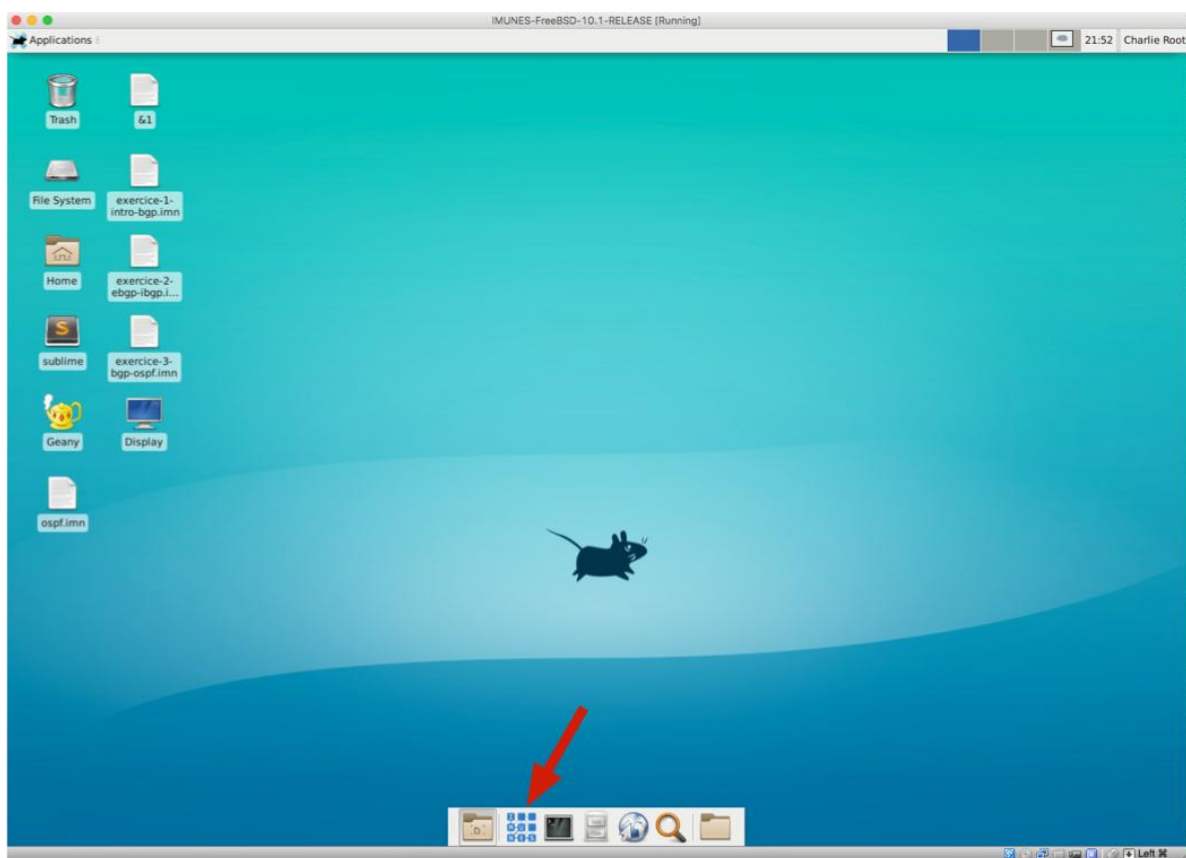
Once the import is complete, click Start.



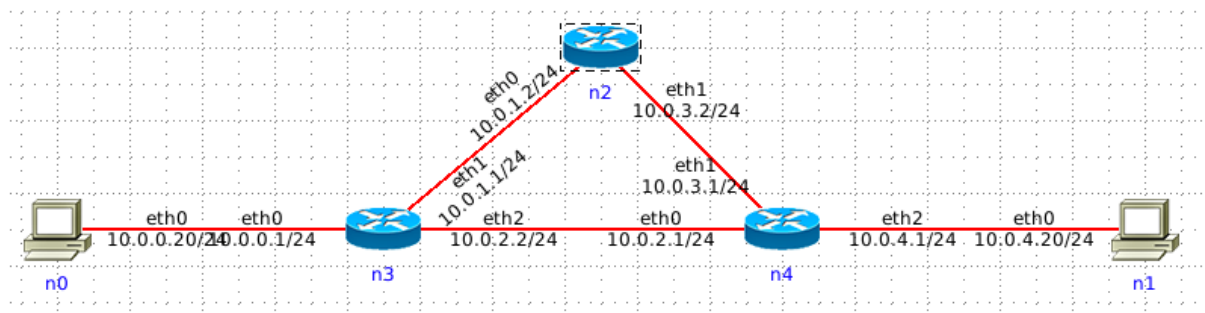
Once you have completed this step, you will end up on the desktop of a BSD system. The keyboard configured by default is a French keyboard (AZERTY). If you want to change to a French or US Apple keyboard, click on the flag as shown below.



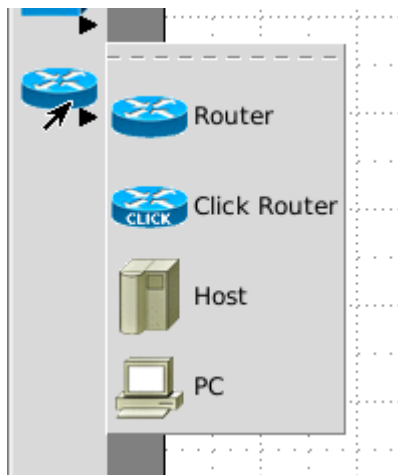
You will then just have to click on the IMUNES icon in the bar at the bottom of the screen to launch the software and start creating a network.



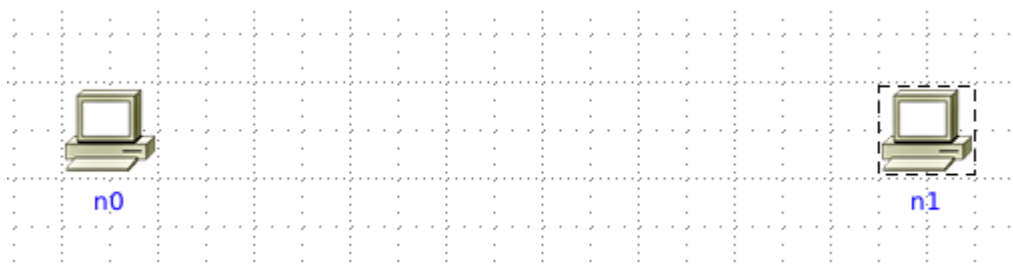
Once IMUNES has been launched, create your first network made up of two PCs and three routers interconnected by communication links.



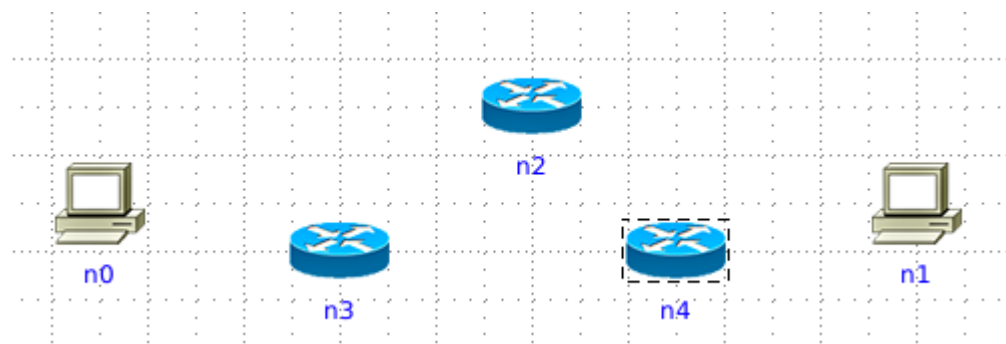
Let's start by creating terminals. To do this, click on the icon representing a router (round) in the left bar. In the menu that appears, select the "PC" tool as in the image below.



The icon changes to represent a PC. You just have to click on the canvas to create a new machine. Create two PCs as in the example below.



We will proceed in the same way to create our routers. To position a router, you will have to select, in the same menu as before, the "Router" tool, and position the routers on the canvas as follows.

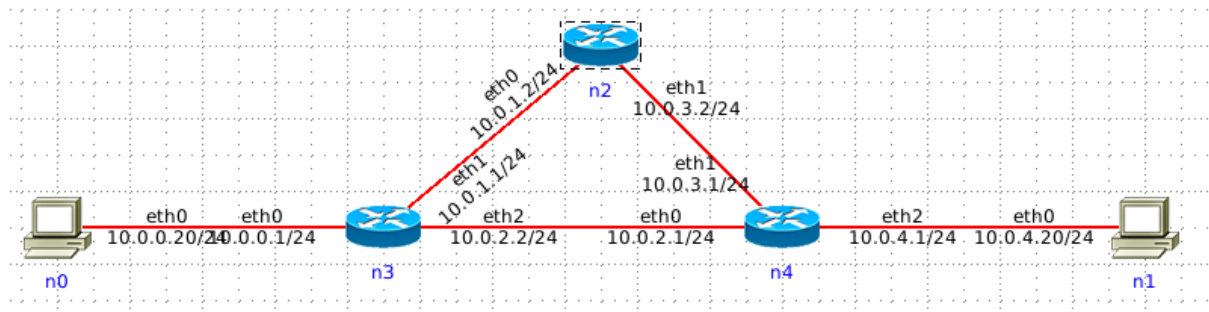


Once all our equipment is created, we will now interconnect them by communication links. To do this, select the "communication link" tool from the left toolbar, the icon of which is as follows:

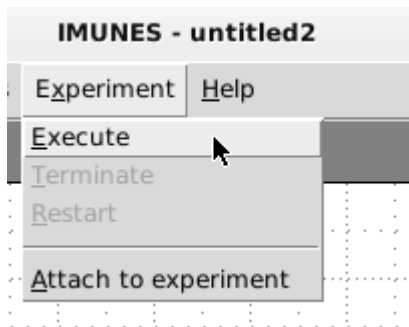


To connect the first PC to a router, all you have to do is drag a line from one to the other while keeping the mouse button pressed. You will notice that if successful, IP addresses appear at both ends of the link created. IMUNES automatically allocates IP addresses to different machines when they are

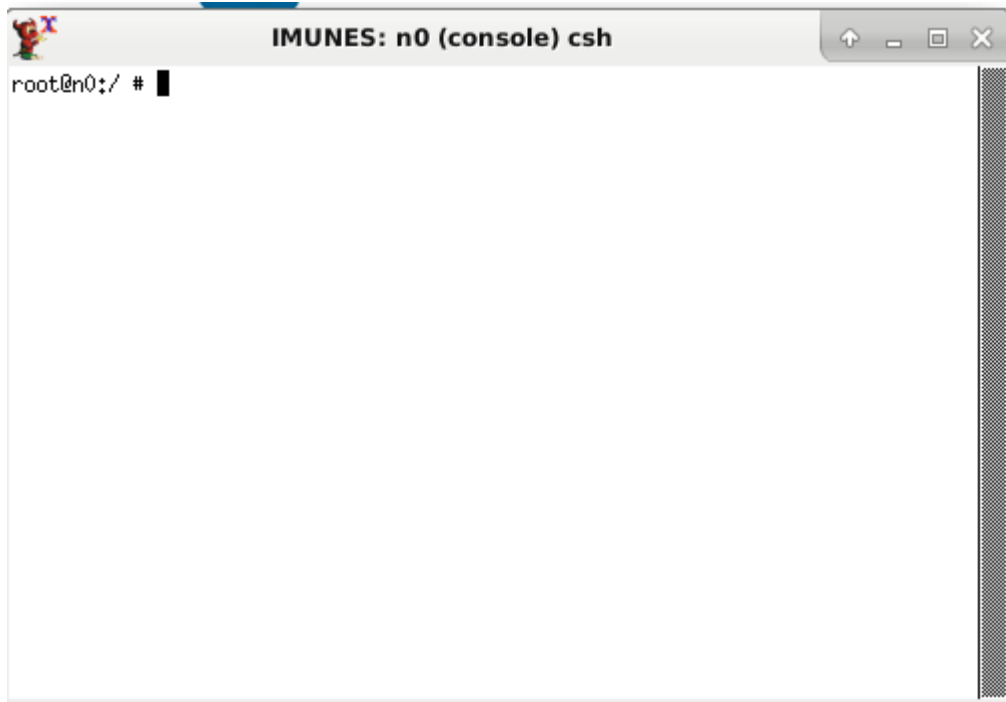
connected. It is of course possible to change these addresses if desired. You can now complete your topology so that it looks like the following image.



We have now told IMUNES which network we would like to create. All that remains is to start the emulator to be able to interact with the machines. To do this, simply go to the "Experiment" menu and select "Execute". The machines start up and you can use them after a few seconds.



To connect to a machine, whether it is a PC or a router, you just need to double-click on its icon when the emulation is running. A terminal window will then appear in which you can enter commands as an administrator.



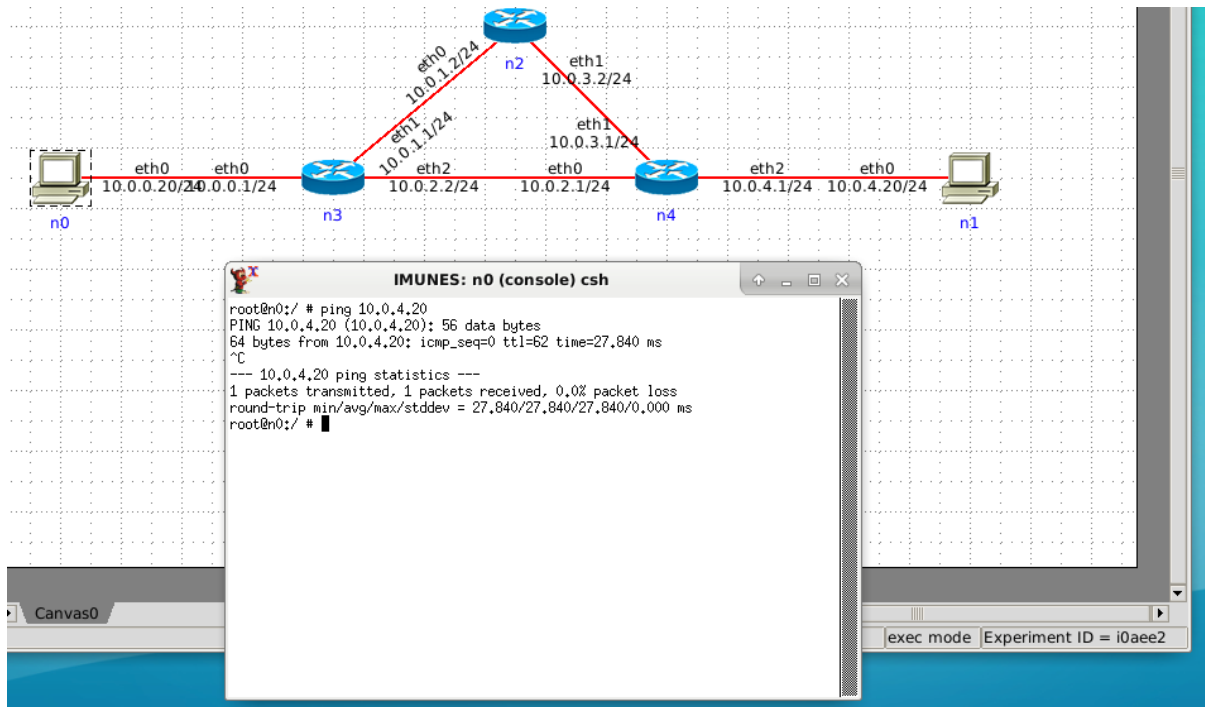
Each of the machines, PC or router, is actually a small Linux machine on which the standard commands for that system run.

Connect to the left PC (here machine n0) and try to contact the right PC using the ping command. You will need to replace the IP address of the destination PC with the real address in your scenario, which you can read on the communication link leading to the PC on the right. In our configuration, we would have to type

```
ping 10.0.4.20
```

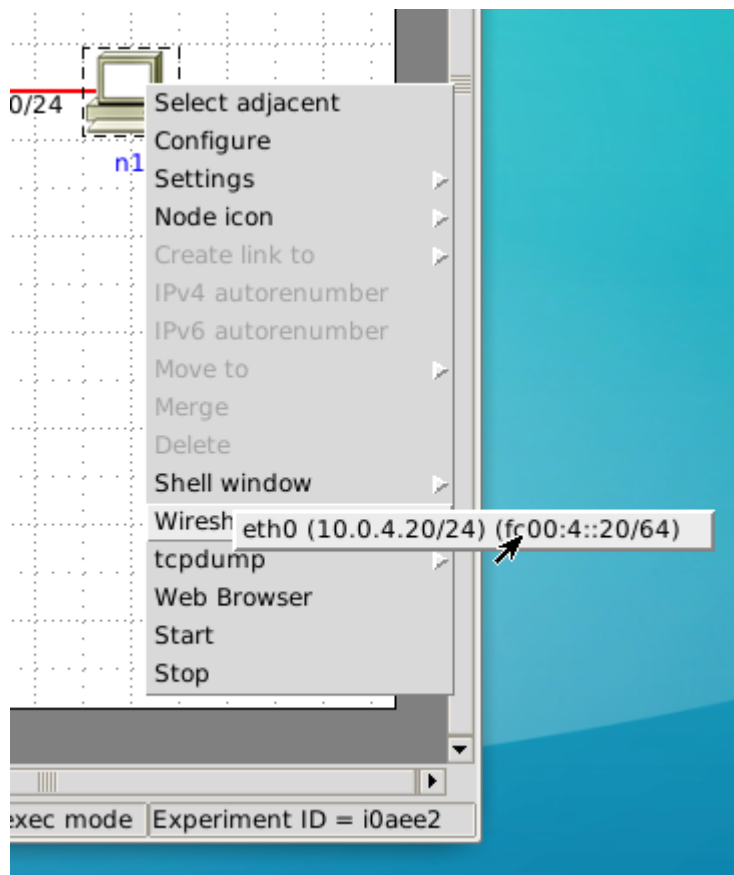
You should normally see successive lines showing you the success of the operation. You can stop the ping command by pressing the Control and C keys simultaneously in the terminal window.

Your network is working: the routers have automatically determined a route connecting the two PCs.

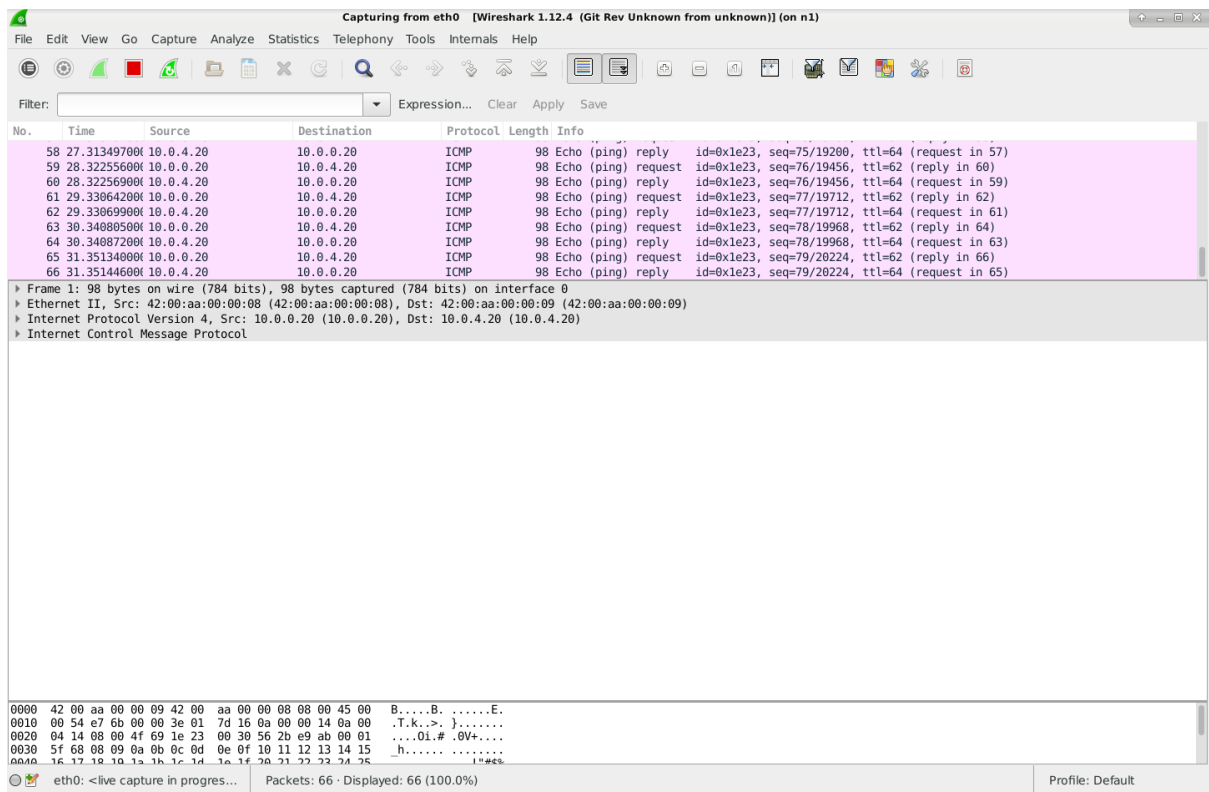


In the last step of getting started with IMUNES, we will examine the traffic on a communication link. To do this, open a window on the left PC and type, as before, the ping command followed by the IP address of the right PC. You won't stop it this time around.

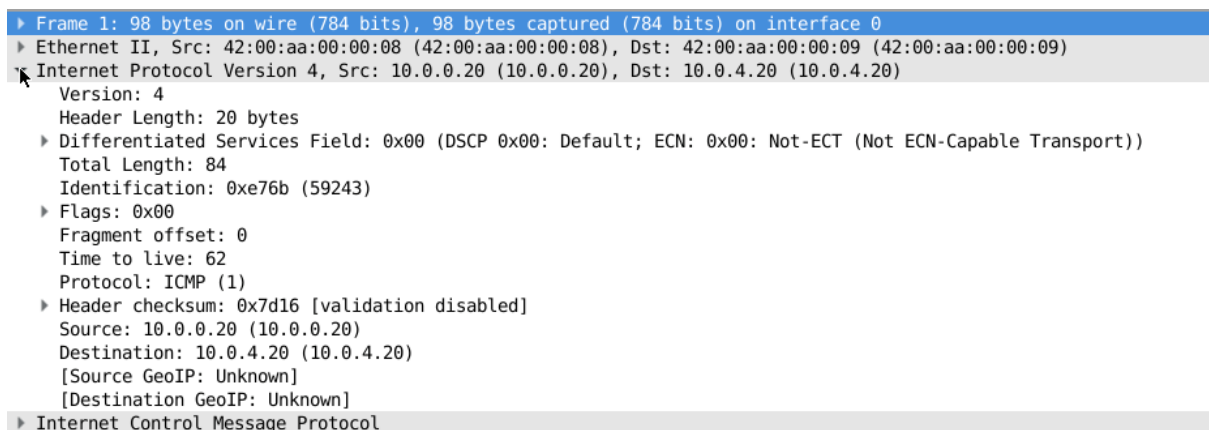
Then right click on the right PC icon. A contextual menu then appears. Select the "Wireshark" submenu then the "eth0" item which corresponds to the first (and only) Ethernet interface of this machine.



A tool for capturing and analyzing frames then starts up: Wireshark. Wireshark will show you the succession of packets arriving or departing from the interface you selected. You may need to resize the window or its parts (by clicking on the edges of the frames in the window) to see the information that interests you.



The window has several areas. Below the toolbar you will find a list of packages that should constantly update in our example. When you select a package from this list, the frame below allows you to see the details of the package: its headers and body. You can click on any of the triangles to the left of a header to display the different fields and their values. For example, in the image below, we have displayed the detail of the IPv4 header of one of the packets generated by the "ping" tool.



Wireshark is a very powerful tool for analyzing what is going on in a network. We encourage you to explore its possibilities and find a tutorial if you are not familiar with this kind of tools.